open data | data on request | closed data | preprint | reanalysis on open data | our reproduction

(a) Maximum vocabulary size used in the 32 English communication-neuroprostheses studies

vocabulary size

71,400 word forms (Brysbaert et al., 2016)
42,000 lemmas (Brysbaert et al., 2016)
11,100 word families (Brysbaert et al., 2016)

6 | 8 | 5–10 | 50 | 178 | 1,024 | 1,024 | 1,152 | 100k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 130k | 130k | open | open | open | open

Angrick et al., 2024 (CC01) | Wandelt et al., 2024 (P1/P2) | Brosler et al., 2026 (B1r/B3/B6) | Moses et al., 2021 (B1) | Silva et al., 2024b (B1) | Metzger et al., 2023 (B3) | Littlejohn et al., 2025 (B3) | Metzger et al., 2022 (B1) | Feghhi et al., 2026 (T12/T15) | Willett et al., 2023 (T12) | Card et al., 2024 (T15) | Feng et al., 2024 (T12) | Kunz et al., 2025 (T12/T15/T16) | Li et al., 2025 (T12) | Feghhi et al., 2025 (T12) | Levin et al., 2026 (T12/T15/T16) | Y. Zhang et al., 2026 (T12/T15) | Nason-Tomaszewski et al., 2026 (T16) | Boccato et al., 2026 (T12/T15/T16/T17) | Jude et al., 2026a (T17/T18) | Jude et al., 2026b (T17) | Yoon et al., 2026 (T12) | Card et al., 2026 (T15) | Fogg et al., 2026 (T12/T15/T16/T17/T21/T22) | Z. Zhang et al., 2026 (T12/T15) | Pouratian et al., 2026 (V1/V2/V3) | Fan et al., 2023 (T5) | Karpowicz et al., 2024 (T5) | Willett et al., 2021 (T5) | Wairagkar et al., 2025 (T15) | Levin et al., 2026 (T5/T12/T18) | Wairagkar et al., 2026 (T15)

Filtered to 8 open-data studies exceeding native-speaker vocabulary (>71,400 word forms)

(b) Open-sourced dataset size

released sentences

10 d/sess 95 blks span 253 d: 1,000
24 d/sess 275 blks span 120 d: 9,680
21 d/sess 105 blks span 287 d: 1,315
45 d/sess 265 blks span 612 d: 9,498
37 d/sess blks n/a span 171 d: 8,489
1 d/sess 12 blks span 1 d: 500
3 d/sess 18 blks span 309 d: 720
2 d/sess 14 blks span 234 d: 450
8 d/sess 46 blks span 89 d: 668
9 d/sess 98 blks span 119 d: 1,033
13 d/sess 96 blks span 141 d: 3,674

Willett et al., 2021 (T5) | Willett et al., 2023 (T12) | Fan et al., 2023 (T5) | ★ Card et al., 2024 (T15) | Wairagkar et al., 2025 (T15) | Kunz et al., 2025 (T12) | Kunz et al., 2025 (T15) | Kunz et al., 2025 (T16) | ★ Jude et al., 2026a (T17) | ★ Jude et al., 2026a (T18) | Jude et al., 2026b (T17)

(c) Word error rate (online)

★ 2 studies satisfy: Open Data, Online, Vocab > 71.4k, WER < 12.8%, Vocal paralysis △

† recomputed from released data (decode outputs / trial timing / Source Data files)
‡ digitized from the paper's figure markers
error bar is calculated across days

word error rate (%)

12.8% 14-dataset average (Radford et al., 2022)
5.8% Switchboard (Xiong et al., 2017)
2.5% LibriSpeech (Radford et al., 2022)

vocab open mean±SD† 5 of 10 d: 25.1±8.6
vocab 125k mean±SD† 5 of 24 d: 23.9±1.8
vocab 130k mean±SD† 10 of 21 d: 4.0±1.2
★ vocab 125k mean±SD 14 of 45 d: 4.9±3.3
vocab open listener median 8 of 37 d: 43.8
vocab 50-w mean±SD† 1 of 1 d: 23.7±0.0
vocab 125k mean±SD† 2 of 3 d: 51.3±33.1
vocab 125k mean±SD† 1 of 2 d: 46.6±0.0
★ vocab 125k mean±SD† 6 of 8 d: 12.5±2.5
★ vocab 125k mean±SD† 2 of 9 d: 3.4±0.8
vocab 125k mean±SD† 4 of 13 d: 66.5±18.1

Willett et al., 2021 (T5) | Willett et al., 2023 (T12) | Fan et al., 2023 (T5) | ★ Card et al., 2024 (T15) | Wairagkar et al., 2025 (T15) | Kunz et al., 2025 (T12) | Kunz et al., 2025 (T15) | Kunz et al., 2025 (T16) | ★ Jude et al., 2026a (T17) | ★ Jude et al., 2026a (T18) | Jude et al., 2026b (T17)

(d) Communication speed (online)

n/r: speed not reported

† recomputed from released data (decode outputs / trial timing / Source Data files)
‡ digitized from the paper's figure markers
error bar is calculated across days

words per minute

160 wpm natural conversation (Card et al., 2024)
51.6 wpm keyboard typing (Dhakal et al., 2018)
36.2 wpm smartphone typing (Palin et al., 2019)
25.1 wpm handwriting (Kaifosh et al., 2025)
5–10 wpm eye typing (Hamid & Kristensson, 2024)
8 wpm point-and-click BCI (Willett et al., 2021)

vocab open mean±SD† 5 of 10 d: 15.4±2.2
vocab 125k mean±SD‡ 5 of 24 d: 65.3±0.9
vocab 130k mean±SD 15 eval d 11 d (6/6/5/5 d): 14.6±2.2
vocab 125k mean±SD 15 of 45 d: 32.5±2.7
n/r
n/r
n/r
n/r
vocab 125k mean±SD† 6 of 8 d: 6.8±1.8
vocab 125k mean±SD† 2 of 9 d: 14.0±4.5
vocab 125k mean±SD† 8 of 13 d: 34.1±7.7

Willett et al., 2021 (T5) | Willett et al., 2023 (T12) | Fan et al., 2023 (T5) | ★ Card et al., 2024 (T15) | Wairagkar et al., 2025 (T15) | Kunz et al., 2025 (T12) | Kunz et al., 2025 (T15) | Kunz et al., 2025 (T16) | ★ Jude et al., 2026a (T17) | ★ Jude et al., 2026a (T18) | Jude et al., 2026b (T17)

Our reproduction on a 64 GB RAM / RTX 5060 Ti (16 GB) home PC

(e) Reproduction: acoustic / synthesis stage (no LM), grouped by metric: PER / CER / 1−r

our reproduction (16 GB home PC) | paper / benchmark anchor

error bar is calculated across days

Greedy PER / CER / 1−r (%)

greedy phoneme error rate (PER) | character error rate (CER) | 1 − mel r (voice)

23.4±3.2 | 19.7
10.2±4.2 | 9.5±2.8
50-w sess. 61.4±0.0 | 24.0±0.0
3-sess. subset 59.7±8.4 | 54.0±33.1
2-sess. subset 61.5±12.1 | 52.0±0.0
40.0±9.2 | 34.0±6.2
4.7±1.9 | 2.8±1.4
seed: 2.65 6.2±3.3 | 6.2±2.3
best 8.2% 19.4±5.8 | 18.6±5.8
best 2.6%; <12% 7.7±2.8 | 12.0±2.4
r: 0.69 vs 0.83 31.4±5.2 | 17.0±4.0

Willett et al., 2023 (T12) | Card et al., 2024 (T15) | Kunz et al., 2025 (T12) | Kunz et al., 2025 (T15) | Kunz et al., 2025 (T16) | Jude et al., 2026b (T17) | Willett et al., 2021 (T5) | Fan et al., 2023 (T5, CORP v6) | Jude et al., 2026a (T17) | Jude et al., 2026a (T18) | Wairagkar et al., 2025 (T15)

(f) Reproduction: full-pipeline WER, one pair per panel-(c) bar

our reproduction (16 GB home PC) | paper / benchmark anchor

error bar is calculated across days

Word error rate (%)

bigram 63-tr. pool 10.0±3.6 | 3.6±3.2
day med. 5-gram 23.6±5.2 | 23.8±4.0
recal. arm 10 sess. 4.1±1.2 | 6.2±2.3
41-s pool biGRU-48k 6.2±6.5 | 6.7±6.6
WER n/r (voice; see e)
50-w sess. gap attr. 94.3±0.0 | 24.0±0.0
125k gap attr. 78.1±0.6 | 54.0±33.1
125k gap attr. 82.4±0.0 | 52.0±0.0
best 4.6% 11.4±6.8 | 12.5±2.5
best 0.0%; <7% 6.7±3.7 | 7.0±2.4
conv. day-68 44.4±16.0 | 50.0±18.1

Willett et al., 2021 (T5) | Willett et al., 2023 (T12) | Fan et al., 2023 (T5) | Card et al., 2024 (T15) | Wairagkar et al., 2025 (T15) | Kunz et al., 2025 (T12) | Kunz et al., 2025 (T15) | Kunz et al., 2025 (T16) | Jude et al., 2026a (T17) | Jude et al., 2026a (T18) | Jude et al., 2026b (T17)